Qn. Set Code-7(SOT)

Semester: 5th
Programme: B.Tech
Branch: Comp. Engg.

AUTUMN END SEMESTER EXAMINATION-2025

5th Semester B.Tech

DESIGN AND ANALYSIS OF ALGORITHMS

CS30001 / CS 2012

(For 2023 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Which of the following algorithm design paradigms satisfies "Optimal Substructure" property?
   i. Greedy, Divide & Conquer
   ii. Dynamic Programming, Greedy
   iii. Divide and Conquer, Dynamic Programming
   iv. All the above

(b) Which of the algorithm uses the random input to reduce the expected running time or memory usage, but always terminate with a correct result in a bounded amount of time?
   i. Las Vegas Algorithm
   ii. Monte Carlo Algorithm
   iii. Dynamic Algorithm
   iv. None of the above

(c) Express in brief with justification what the following algorithm computes for the matrix (A) of size n x n.

```
Boolean Enigma(A)
FOR i ← 1 to n − 1 DO
    FOR j ← i + 1 to n DO
```

IF $A[i,j] \neq A[j,i]$ THEN
RETURN *false*
RETURN *true*

(d) When a top-down approach of dynamic programming is applied to a problem, it usually _____

i. Decreases both, the time complexity and the space complexity
ii. Decreases the time complexity and increases the space complexity
iii. Increases the time complexity and decreases the space complexity
iv. Increases both, the time complexity and the space complexity

(e) Define Matrix Chain Multiplication Problem.

(f) Consider the following functions:

$f(n) = 3n^{\sqrt{n}}$, $g(n) = 2^{\sqrt{n}\log_2 n}$, $h(n) = n!$

Which of the following is/are true?

i. $h(n) = \Theta(f(n))$
ii. $h(n) = \Theta(g(n))$
iii. $f(n) = \Theta(g(n))$
iv. $g(n) \neq \Theta(f(n))$

(g) Let G be an edge-weighted undirected graph with positive edge weights. Suppose a positive constant α is added to the weight of every edge. Which ONE of the following statements is TRUE about the minimum spanning trees (MSTs) and shortest paths (SPs) in G before and after the edge weight update?

i. Every MST remains an MST, and every SP remains an SP.
ii. MSTs need not remain MSTs, and every SP remains an SP.
iii. Every MST remains an MST, and SPs need not remain SPs.
iv. MSTs need not remain MSTs, and SPs need not remain SPs.

(h) Consider the following C-program fragment in which i, j and n are integer variables.

```
i = n;
for (j = 0; i > 0; i /= 2)  j += i;
```

Let val(j) denotes the value stored in the variable j after termination of the for loop. Find the tightest asymptotic bound for the function val(j).

(i) 05. Consider a decision problems Q. 3CNF–SAT reduces in polynomial time to Q. Then which one of the following is consistent with the above statement?

i. Q is in NP

ii. Q is in NP hard.

iii. Q is in NPC

iv. Q is in P

(j) Define equal length encoding and how is it different from Hoffman encoding?

SECTION-B

2. (a) An array of integers is said to be UNIMODAL if all its entries are distinct and they are in increasing order until its largest element, after which elements are in decreasing order. Design an algorithm to compute the largest element of a UNIMODAL that runs in O(log n). [6]

(b) Solve the following recurrence – [4]

$T(n) = T(n/4) + T(3n/4) + n$; for $n>1$

$T(1) = 1$

3. (a) Given an array, A of size n. Write an algorithm to push one digit number from position 1 to n in order in A. Update all the previously stored numbers from index position i-1 down to 1 while pushing a digit (d) to the index position i ($i=2^k$; $k=>1$) by 10 x A(p) + d (for p= i-1 down to 1). [6]

(b) Use aggregate analysis or accounting method to determine the amortized running cost per operation (both pushing and update) of the algorithm in 4(a). [4]

4. (a) Select(a, n) is a function that returns the largest element position in an unsorted array, a from position 1 to n on a non-deterministic machine with running time O(1). Write an algorithm to sort n elements in an array in sorted order and find the running time. [5]

(b) The characters 'a' to 'h' have the following set of frequencies: {a : 1, b : 1, c : 2, d : 3, e : 5, f : 8, g : 13, h : 21}. A Huffman code is used to represent the characters. Find the character string corresponding to the following encoded bit string – 110111100111010, provided that it is a valid encoded text. [5]

5. (a) Let $S = \{a, b, c, d, e, f, g\}$ be a collection of objects with benefit-weight values given as follows: a: (12, 4), b: (10, 6), c: (8, 5), d: (11, 7), e: (14, 3), f: (7, 1), g: (9, 6). Derive an optimal solution to the 0/1 knapsack problem for S assuming knapsack can hold objects with total weight 18. [5]

(b) Suggest the data structure(s) used to store the MCST of a graph in the Kruskal's Algorithm. Write an algorithm to display the edges of the MCST. [5]

6. (a) Show the longest common subsequence table, L, for the following two strings: X = "skullandbones", Y = "lullabybabies". Find a longest common subsequence between these strings using a dynamic programming method. [5]

(b) Define NP with respect to Clique Decision Problem (CDC). Write an algorithm to prove CDC belongs to NP. [5]

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